ME 165
Basic Mechanical Engineering

Lecture 7

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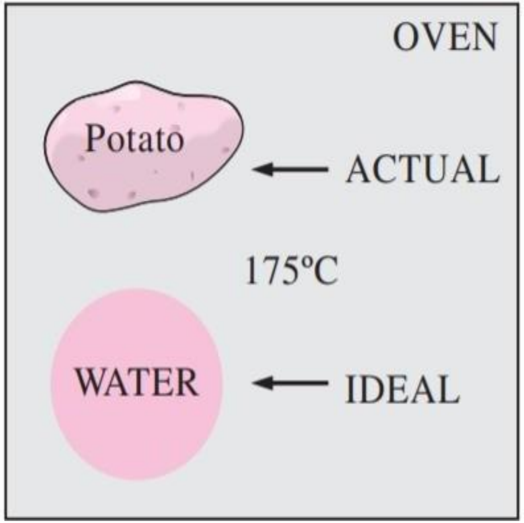
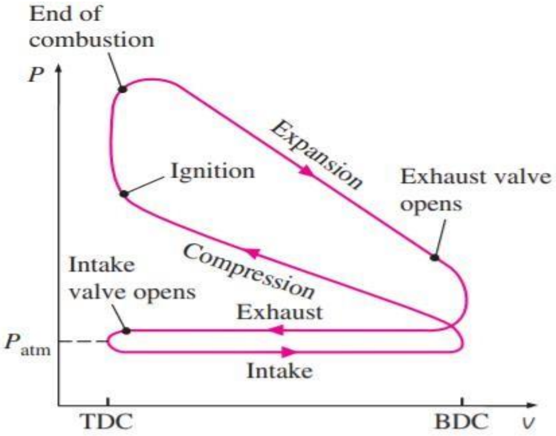
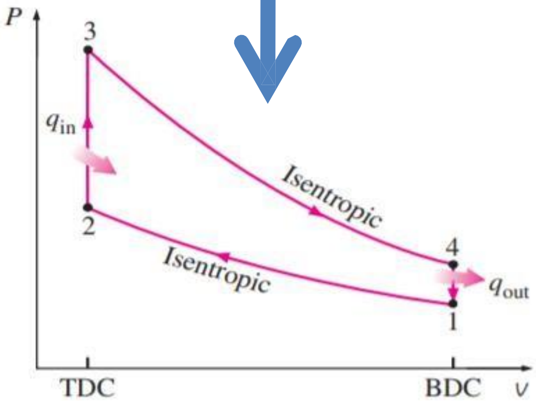
Air Standard Cycle

Reference Book: Thermodynamics: An Engineering Approach, Yunus A. Cengel, Michael A. Boles
Chapter -9 (Initial Portion)

Necessity of an Ideal Cycle







To make an analytical study of a cycle feasible, we have to keep the complexities at a manageable level and utilize some idealizations.

Air - Standard Assumptions

- The accurate study and analysis of I.C. engine processes is very complicated. To simplify the theoretical study "***Standard Air Cycles***" are introduced

Assumptions

- The working fluid is air, which continuously circulates in a closed loop and always behaves as an ideal gas.
- All the processes that make up the cycle are internally reversible.
- The combustion process is replaced by a heat-addition process from an external source.
- The exhaust process is replaced by a heat-rejection process that restores the working fluid to its initial state.

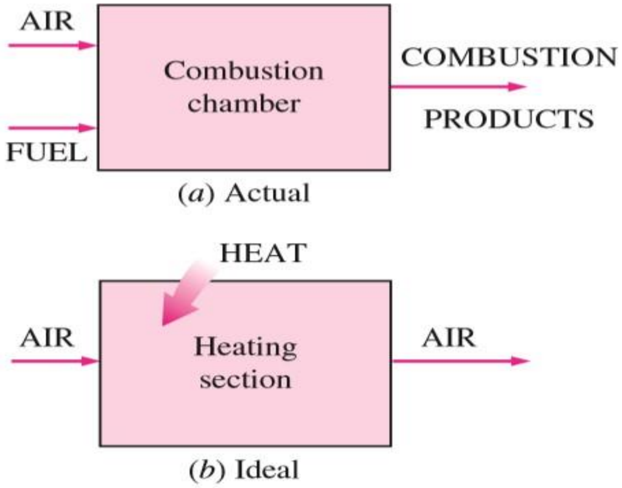


FIGURE 9–9

The combustion process is replaced by a heat-addition process in ideal cycles.

Air - Standard Cycles

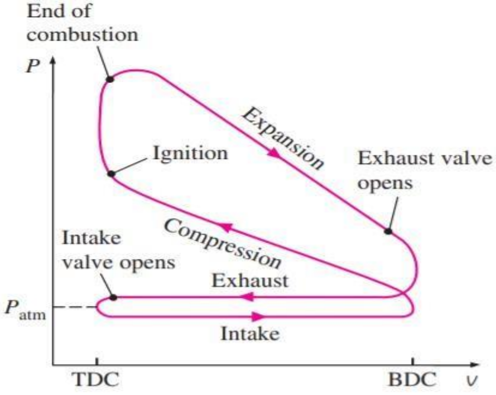
Thermodynamic cycles that adhere to air-standard assumptions are –

- Otto Cycle or Petrol Cycle (for SI Engines)
- Diesel Cycle (for CI Engines)

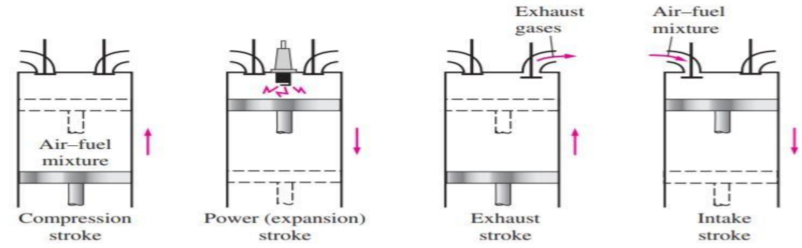
Otto Cycle:

The Otto cycle is the ideal cycle for spark-ignition reciprocating engines. It is named after Nikolaus A. Otto, who built a successful four-stroke engine in 1876 in Germany using the cycle proposed by Frenchman Beau de Rochas in 1862.

Air Standard Otto (constant volume) or Petrol Cycle

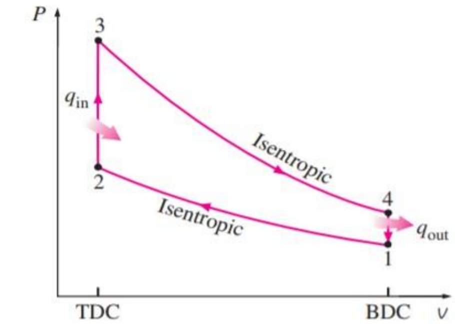


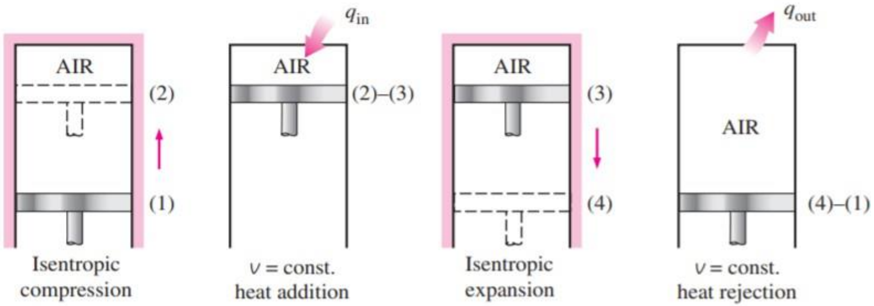
Actual Cycle



(*a*) Actual four-stroke spark-ignition engine

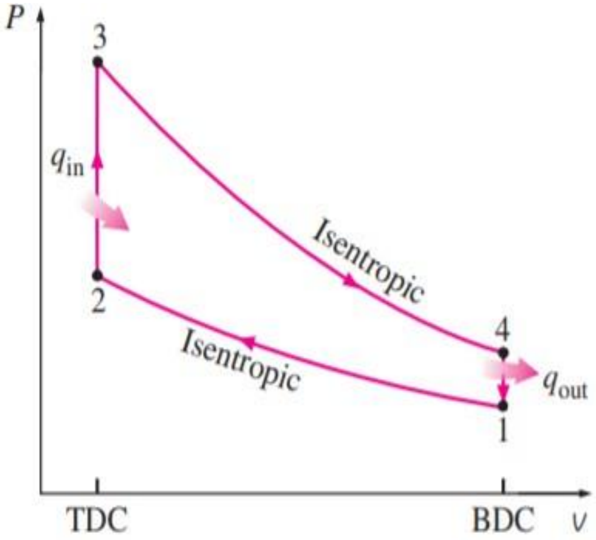
Ideal Cycle

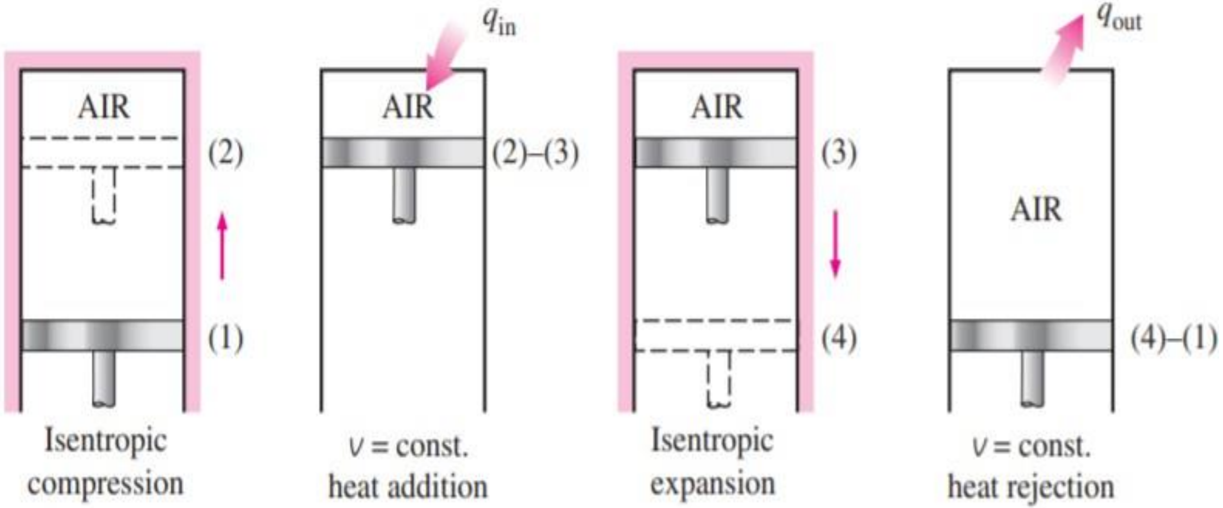




(*b*) Ideal Otto cycle

Air Standard Otto (constant volume) or Petrol Cycle





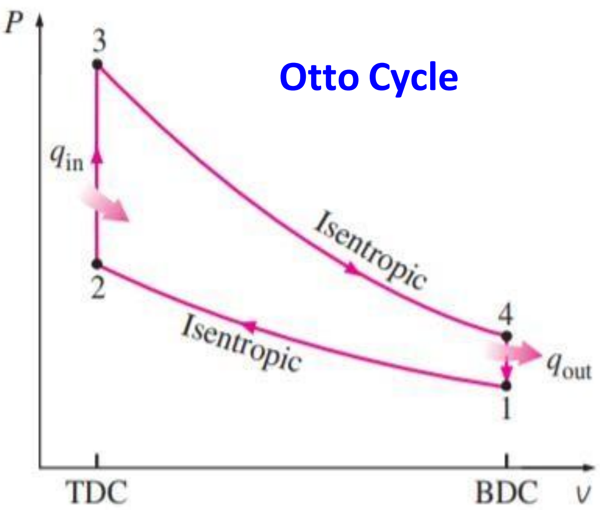
(b) Ideal Otto cycle

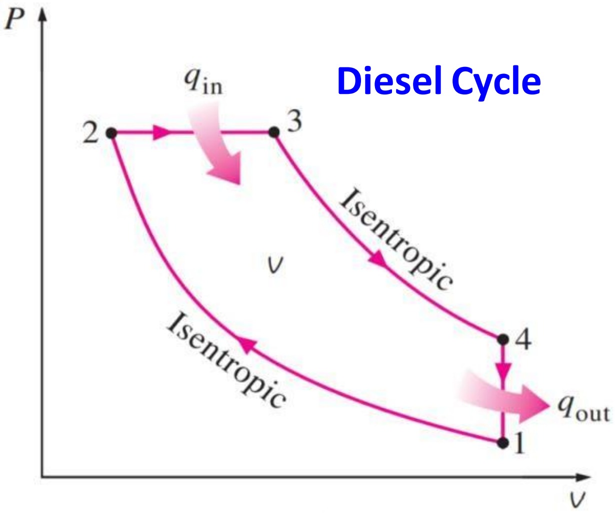
- 1-2 Isentropic compression
- 2-3 Constant-volume heat addition
- 3-4 Isentropic expansion
- 4-1 Constant-volume heat rejection

Why constant volume approximation in Otto Cycle?

- The air-standard Otto cycle assumes the **heat addition to be occurring instantaneously while the piston is at TDC**.
- Therefore, in petrol cycle, heat addition will take place at constant volume.
- This is due to the fact that the pressure inside the cylinder will rise rapidly during the combustion period as soon as ignition of air-fuel mixture is started by a spark plug.

Air Standard Diesel (Constant Pressure) Cycle





- Air standard Diesel Cycle assumes the heat addition occurs during a constant pressure process that starts with the piston at TDC.
- **Process 1-2: Isentropic Compression** (Piston: BDC → TDC)
- **Process 2-3: Constant Pressure Heat Addition** (1st part of Power Stroke). (Piston: TDC → BDC)
- **Process 3-4: Isentropic Expansion** (Remainder of Power Stroke). (Piston: TDC → BDC)
- **Process 4-1: Constant Volume Heat Rejection** (Piston: at BDC)
- This cycle is the theoretical cycle for compression-ignition or diesel engine.

Why constant-pressure heat-addition process approximation?

- The fuel injection process in diesel engines starts when the piston approaches TDC and continues during the first part of the power stroke.
- Therefore, <u>the combustion process in these engines takes place over a longer interval.</u>
- Because of this longer duration, the combustion process in the ideal Diesel cycle is approximated as a constant-pressure heat-addition process.

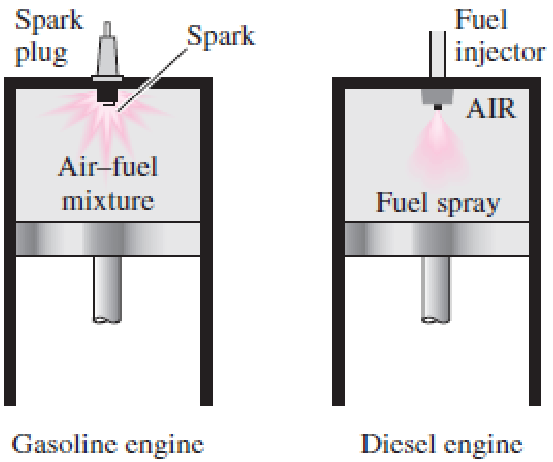


FIGURE 9–20
In diesel engines, the spark plug is replaced by a fuel injector, and only air is compressed during the compression process.

Isentropic Process

- **Entropy** is the measure of the disorder of a system.
- The term "isentropic" means constant entropy. A process during which the entropy remains constant is called an isentropic process

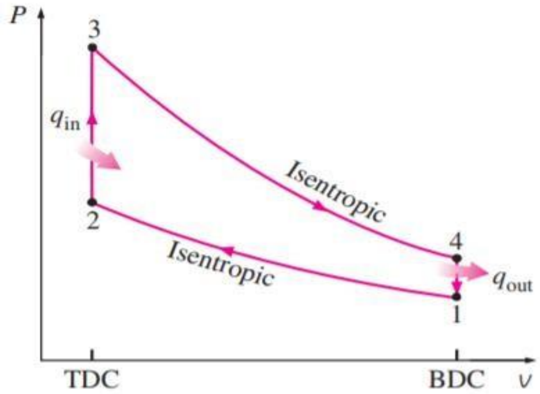
$$PV^{K} = Constant$$
$$TV^{K-1} = Constant$$
$$TP^{\frac{1-k}{k}} = Constant$$
$$k = \frac{C_p}{C_v}$$
$$k = specific\ heat\ ratio$$

Entropy, the measure of a system's thermal energy per unit temperature that is unavailable for doing useful work. Because work is obtained from ordered molecular motion, the amount of **entropy** is also a measure of the molecular disorder, or randomness, of a system.

- **Cv** for a gas is the change in internal energy (U) of a system with respect to change in temperature at a fixed volume of the system i.e. $C_v = \left(\frac{\partial U}{\partial T}\right)_V$

- **Cp** for a gas is the change in the enthalpy (H) of the system with respect to change in temperature at a fixed pressure of the system i.e $C_p = \left(\frac{\partial H}{\partial T}\right)_P$

Efficiency of Otto Cycle



- No work is involved during the two heat transfer processes since both take place at constant volume. Therefore, heat transfer to and from the working fluid can be expressed as

$$q_{in} = u_3 - u_2 = c_v(T_3 - T_2) \qquad q_{out} = u_4 - u_1 = c_v(T_4 - T_1)$$

- The thermal efficiency of the ideal Otto cycle under the cold air standard assumptions becomes

$$\eta_{th,Otto} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{T_4 - T_1}{T_3 - T_2} = 1 - \frac{T_1(T_4/T_1 - 1)}{T_2(T_3/T_2 - 1)}$$

- Processes 1-2 and 3-4 are isentropic, and v2 = v3 and v4 = v1. Thus,

$$\frac{T_1}{T_2} = \left(\frac{v_2}{v_1}\right)^{k-1} = \left(\frac{v_3}{v_4}\right)^{k-1} = \frac{T_4}{T_3} \rightarrow r = \frac{V_{max}}{V_{min}} = \frac{V_1}{V_2} = \frac{v_1}{v_2} \rightarrow \eta_{th,Otto} = 1 - \frac{1}{r^{k-1}}$$

Efficiency of Otto Cycle

- η increased by increasing r,
- η increased by increasing k,
- In modern petrol engines (r) reaches a value of 12
- The specific heat ratio k, and thus the thermal efficiency of the ideal Otto cycle, decreases as the molecules of the working fluid get larger.
- At room temperature it is 1.4 for air, 1.3 for carbon dioxide, and 1.2 for ethane.
- The working fluid in actual engines contains larger molecules such as carbon dioxide, and the specific heat ratio decreases with temperature, which is one of the reasons that the actual cycles have lower thermal efficiencies than the ideal Otto cycle. The thermal efficiencies of actual spark- ignition engines range from about 25 to 30 percent.

$$\eta_{\text{th,Otto}} = 1 - \frac{1}{r^{k-1}}$$

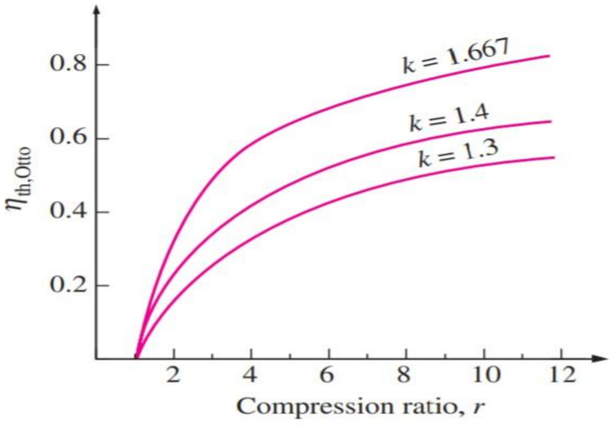


FIGURE 9–18

The thermal efficiency of the Otto cycle increases with the specific heat ratio k of the working fluid.

Thermal Efficiency of the Ideal Diesel Cycle

$$q_{in} = P_2(v_3 - v_2) + (u_3 - u_2)$$
$$= h_3 - h_2 = c_p(T_3 - T_2)$$

$$q_{out} = u_4 - u_1 = c_v(T_4 - T_1)$$

$$\eta_{th,Diesel} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{c_v(T_4 - T_1)}{c_p(T_3 - T_2)}$$

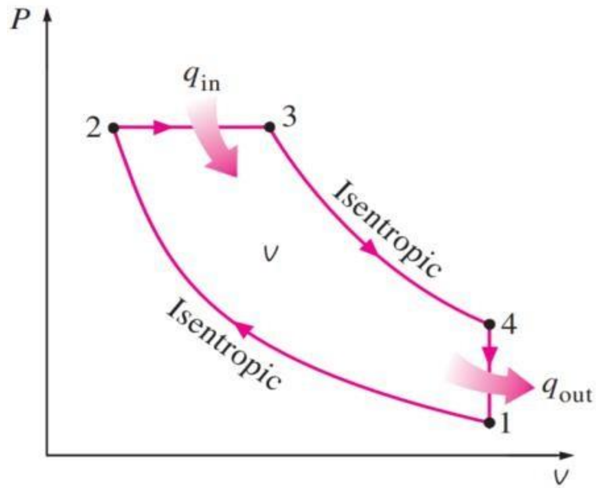
$$\therefore \eta_{th,Diesel} = 1 - \frac{T_4 - T_1}{k(T_3 - T_2)} = 1 - \frac{T_1(T_4/T_1 - 1)}{kT_2(T_3/T_2 - 1)}$$

$Now, Cutoff\ rato, r_c = \dfrac{V_3}{V_2}$ and $Compression\ ratio, r = \dfrac{V_1}{V_2}$

$Using, i)\ TV^{K-1} = Constant\ for\ isentropic\ process\ (1 - 2\ \&\ 3 - 4)$

$ii)\ \dfrac{V}{T} = Constant\ for\ isobaric\ process, iii)\ r_c$ and $iv)\ r$

$$We\ get, \eta_{th,Diesel} = 1 - \frac{1}{r^{k-1}}\left[\frac{r_c^k - 1}{k(r_c - 1)}\right]$$



(a) P- v diagram

Thermal Efficiency Comparison

<u>Diesel Cycle</u>

$$\eta_{\text{th,Diesel}} = 1 - \frac{1}{r^{k-1}}\left[\frac{r_c^k - 1}{k(r_c - 1)}\right]$$

<u>Otto Cycle</u>

$$\eta_{\text{th,Otto}} = 1 - \frac{1}{r^{k-1}}$$

- The efficiency of a Diesel cycle differs from the efficiency of an Otto cycle by the quantity in the brackets. This quantity is always greater than 1. Therefore, when both cycles operate on the same compression ratio, $\eta_{th,otto} > \eta_{th,Diesel}$
- Diesel engines operate at much higher compression ratios and thus are usually more efficient than the spark-ignition (gasoline) engines.
- Thermal efficiencies of large diesel engines range from about 35 to 40 percent.

Mean Effective Pressure

- It is a **theoretical parameter** used to measure the performance of an internal combustion engine (ICE).
- The mean effective pressure can be regarded as an average pressure in the cylinder for a complete engine cycle. By definition, mean effective pressure is the ratio between the work and engine displacement:

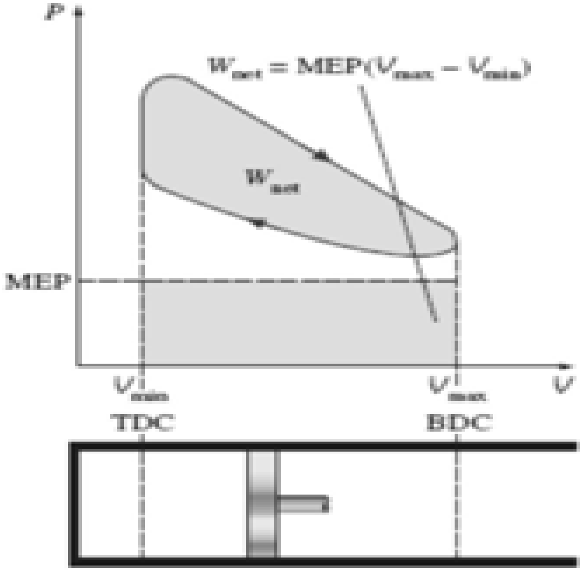
$$MEP = W/V_d \ [Pa]$$

W [J] – work performed in a complete engine cycle

V_d[m³] – engine (cylinder) displacement

W_{net} = MEP * Piston area * Stroke

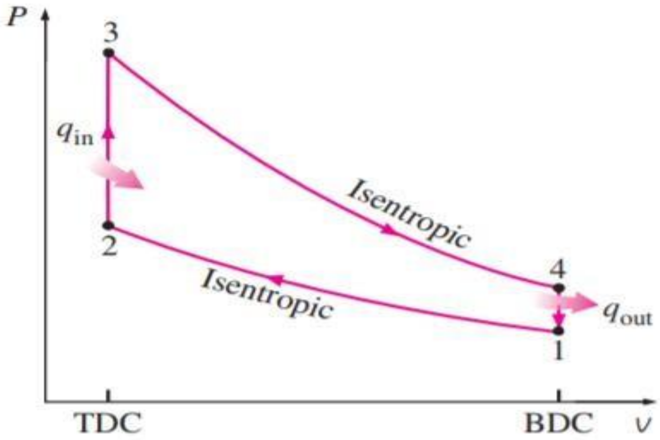
= MEP * Displacement Volume



- The MEP can be used as a parameter to compare the performance of reciprocating engines of equal size.
- The engine with a larger MEP delivers more net work per cycle and thus better performance.

Problem 1

In an air standard Otto cycle, the compression begins at 35 °C and 0.1 MPa and the compression ratio is 7. The maximum temperature of the cycle is 1100°C. Find (a) the temperature at various points in the cycle, (b) the heat supplied per kg of air, (c) work done per kg of air, (d) the cycle efficiency



T_1=35°C=308 K
P_1=0.1MPa
T_3=1100°C=1373K
$r=v_1/v_2=7$

(a) At 300 K

Gas	Formula	Gas constant, R kJ/kg·K	c_p kJ/kg·K	c_v kJ/kg·K	k
Air	—	0.2870	1.005	0.718	1.400

Problem 1

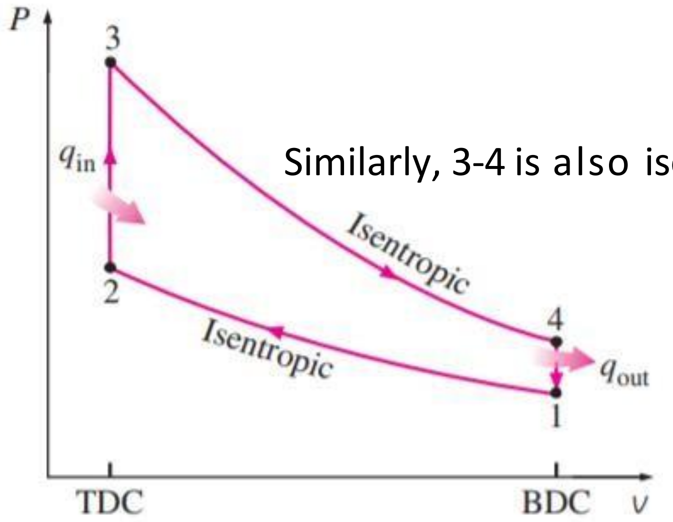
Since process, 1-2 is isentropic,

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^k = 7^{1.4} = 15.24 \qquad P_2=1524 \text{ kPa}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1} = 7^{1.4-1} = 2.178 \qquad T_2=670.8 \text{ K}$$

Similarly, 3-4 is also isentropic,

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{k-1} \qquad T_4=631 \text{ K}$$



Heat rejected, $Q_{out}= C_v (T_4-T_1) = 231.44$ kJ/kg

Heat in, $Q_{in}= C_v (T_3-T_2) = 504.8$ kJ/kg

The net work output, $W_{net} = Q_{in} - Q_{out} = 272.74$ kJ/kg

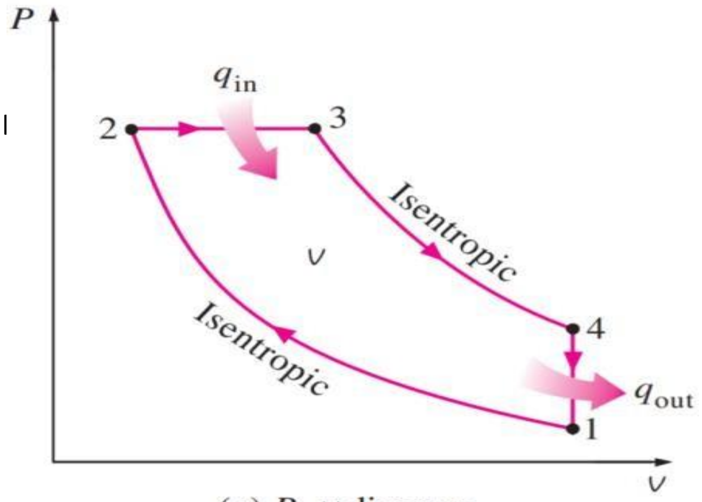
Thermal efficiency, $\eta_{th,otto} = W_{net} / Q_{in} = 0.54 = 54$ %

Otto cycle thermal efficiency, $\eta_{th,otto} = 1 - 1/r^{k-1} = 0.54$ or 54 %

Problem 2

In a Diesel cycle, Compression begins at 0.1 MPa, 40°C and the compression ratio is 15. The heat added is 1.675 MJ/kg and the compression ratio is 15. Find

(a) the maximum temperature in the cycle,
(b) the cut- off ratio
(c) the temperature at the end of the isentropic expansion
(d) work done per kg of air
(e) the cycle efficiency
(f) the MEP of the cycle.



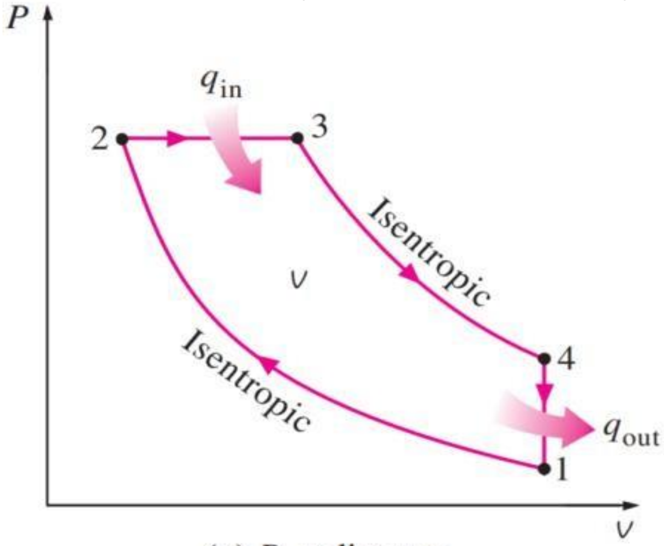
(a) P- v diagram

T_1=40 °C=313 K
P_1=0.1MPa
Q_{in}=1675 MJ/kg
$r=v_1/v_2$=15

(a) At 300 K

Gas	Formula	Gas constant, R kJ/kg·K	c_p kJ/kg·K	c_v kJ/kg·K	k
Air	—	0.2870	1.005	0.718	1.400

Problem 2



(a) P- v diagram

Since process, 1-2 is isentropic, $\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^k$ P_2=4431 kPa $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1}$ T_2=924.66 K

$Q_{in} = C_p (T_3 - T_2)$ $T_3 = T_{max} = 2591.33$ K

$$\frac{P_2 v_2}{T_2} = \frac{P_3 v_3}{T_3} \qquad \frac{v_2}{v_3} = r_c = 2.8$$

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{k-1}$$ T_4=1325.37 K

Heat rejected, $Q_{out} = C_v (T_4 - T_1)$ = 726.88 KJ/kg

The net work output, $W_{net} = Q_{in} - Q_{out}$ = 948.12 kJ/kg

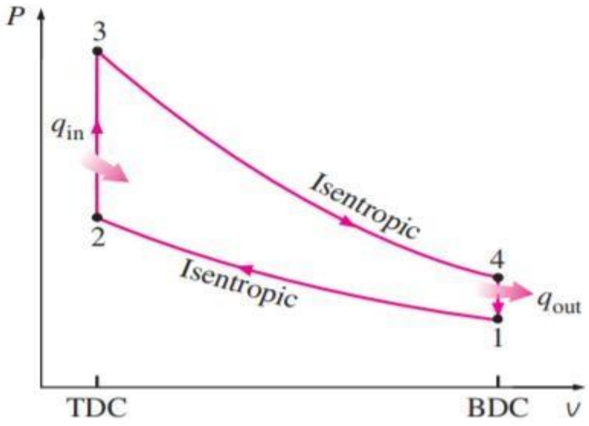
Thermal efficiency, $\eta_{th} = W_{net} / Q_{in}$=0.566=56.6 %

The mean effective pressure = $W_{net} / (v_1 - v_2)$ = 1131. 4 KPa

Problem 3

The compression ratio in an air-standard Otto cycle is 10. At the beginning of the compression stoke, the pressure is 0.1 MPa and the temperature is 15°C. The heat transfer to the air per cycle is 1800 kJ/kg air. Determine

1. The pressure and temperature at the end of each process of the cycle.
2. The thermal efficiency.
3. The mean effective pressure.



Problem 4

An ideal Diesel cycle with air as the working fluid has a compression ratio of 18 and a cutoff ratio of 2. At the beginning of the compression process, the working fluid is at 14.7 psia, 80°F, and 117 in^3. Utilizing the cold-air-standard assumptions, determine (*a*) the temperature and pressure of air at the end of each process, (*b*) the net work output and the thermal efficiency, and (*c*) the mean effective pressure.

Acknowledgement

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